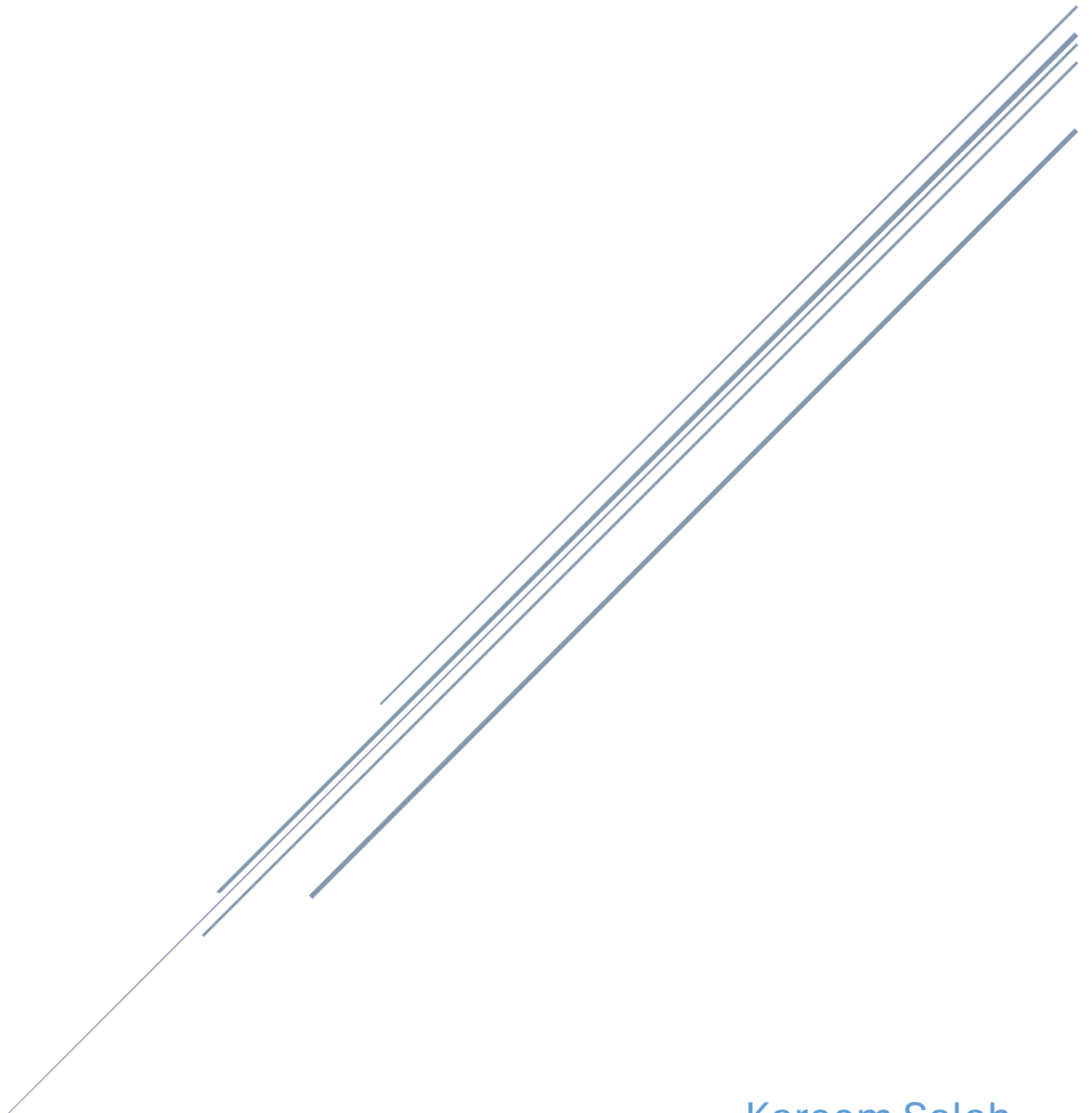


FIFA 22

Documentation



Kareem Salah

FIFA 22 Players Overall Rating Prediction

Comprehensive Project Documentation

Table of Contents

1. [Project Overview](#)
 2. [Dataset Description](#)
 3. [Project Objectives](#)
 4. [Data Processing & Cleaning](#)
 5. [Exploratory Data Analysis \(EDA\)](#)
 6. [Feature Engineering](#)
 7. [Machine Learning Models](#)
 8. [Model Performance & Results](#)
 9. [Key Insights](#)
 10. [Deployment](#)
-

Project Overview

This project is a comprehensive machine learning pipeline designed to **predict FIFA 22 player overall ratings** using player attributes, physical characteristics, financial data, and performance metrics. The analysis covers data exploration, preprocessing, feature engineering, and the implementation of 15+ machine learning models.

Dataset: FIFA 22 Official Player Data

Records: 16,710 players

Features: 65 columns

Target Variable: Overall Rating (0-99 scale)

Dataset Description

Dataset Dimensions

- **Shape:** 16,710 rows × 65 columns

- **Player Population:** Professional football players from FIFA 22 video game database

Key Attribute Categories

Player Demographics

- Player ID, Name, Age, Nationality, Club, Position
- Height (cm), Weight (kg), Preferred Foot
- Jersey Number

Financial Attributes

- **Market Value (€):** Estimated player market value
- **Weekly Wage (€):** Current salary
- **Release Clause (€):** Buyout clause value

Performance Ratings

- **Overall Rating:** Current player rating (0-99)
- **Potential:** Maximum rating achievable (0-99)
- **Best Overall Rating:** Highest rating achieved at best position

Technical Skills (0-99 scale)

Offensive: Crossing, Finishing, Short Passing, Volleys, Dribbling, Curve, Long Shots, Ball Control

Defensive: Marking, Standing Tackle, Sliding Tackle, Interceptions, Positioning

Special Attributes: Vision, Composure, Special

Physical Attributes (0-99 scale)

- Acceleration, Sprint Speed, Agility, Balance, Strength, Stamina, Jumping
- Height and Weight relationships

Goalkeeping Attributes (GK only)

- Diving, Handling, Kicking, Positioning, Reflexes

Career Attributes

- International Reputation (1-5 stars)

- Skill Moves (1-5 stars)
 - Weak Foot (1-5 stars)
 - Work Rate (Attacking/Defensive - Low/Medium/High)
 - Body Type Classification
 - Contract Year, Years at Club
-

Project Objectives

The primary goals of this analysis are:

1. **Data Exploration:** Understand distributions and characteristics of FIFA 22 players
 2. **Quality Assessment:** Identify missing values, outliers, and data inconsistencies
 3. **Feature Relationships:** Explore correlations between player attributes
 4. **Feature Engineering:** Create meaningful derived features to enhance model performance
 5. **Predictive Modeling:** Build and compare multiple ML models for rating prediction
 6. **Model Optimization:** Hyperparameter tuning and ensemble methods
 7. **Deployment:** Prepare trained model for production use
-

Data Processing & Cleaning

1. Initial Data Assessment

Missing Values Analysis:

- Marking: 95.3% missing → **Dropped**
- Loaned From: 91.8% missing → **Dropped**
- Photo, Flag, Club Logo, Real Face: Non-essential media → **Dropped**

Low Missing Rates (<2%):

- Volleys, Club, Composure, Curve, Agility, Jumping, Vision, SlidingTackle, Balance, Body Type, Jersey Number, Position, Interceptions, Positioning

- **Action:** Rows with these missing values removed (minimal impact)

2. Data Type Conversions

Financial Columns (Value, Wage, Release Clause):

- Original format: "€123.5M", "€50K"
- Conversion function created to handle M (millions) and K (thousands)
- Converted to numeric (EUR)

Date Columns:

- Joined: String → DateTime
- Contract Valid Until: String → DateTime → Extract Year

Physical Measurements:

- Height: String "180cm" → Integer (cm)
- Weight: String "75kg" → Integer (kg)

Position Column:

- Extracted position from HTML-formatted field
- Cleaned to standard position abbreviations

3. Missing Value Handling

DefensiveAwareness:

- Filled by position-wise median (position-specific imputation)
- Fallback: Overall median for unmapped positions

Release Clause:

- Filled by grouping on Overall rating bins (binned median imputation)
- Preserves relationship between rating and clause value

Joined Date:

- Filled by most common year per club (mode)
- Fallback: Overall median

Contract Year:

- Filled by most common year per club
- Fallback: Overall median

Final Result: 0 null values after processing

4. Outlier Management

Outlier Detection Method: IQR (Interquartile Range)

- Lower Bound = $Q1 - 1.5 \times IQR$
- Upper Bound = $Q3 + 1.5 \times IQR$

Strategy Adopted:

- **Retained outliers** in all features (wage, value, skills)
- **Rationale:** Extreme values represent real players, not data errors
- **Approach:** Used robust models (Tree-based, Ensemble) to handle outliers naturally

Log Transformation Applied:

- Value, Wage, Release Clause, Special $\rightarrow$ $\log_{10}$ transformation
- Reduces right-skewness for better model convergence

Exploratory Data Analysis (EDA)

Univariate Analysis

Age Distribution

- **Range:** 15-44 years
- **Mode:** Late teens to early 30s
- **Key Finding:** Most players concentrated in 20s-30s (prime years)

Overall Rating Distribution

- **Range:** 40-94
- **Mean Peak:** Around 70
- **Pattern:** Slight left skew (more players rated 70-75 than 80+)

Potential Rating Distribution

- **Correlation with Overall:** 0.58 (moderate positive)
- **Pattern:** High potential players typically have high current ratings

Position Distribution

- **27 distinct positions**
- **Top Positions:** CM (Central Midfielder), CB (Centre-Back), ST (Striker)
- **Rarest:** SUB, RES (substitute/reserve classification)

Preferred Foot

- **Right-footed:** ~88% (majority)
- **Left-footed:** ~12% (rarer, slight rating advantage)

International Reputation

- **Distribution:** Heavily skewed toward 1-star (common players)
- **5-star:** Rare superstars (Messi, Ronaldo tier)

Body Type Categories

- **Lean:** Most common
- **Stocky:** Mid-range
- **Athletic:** Elite players

Bivariate Analysis

Age vs Overall Rating

- **Correlation:** +0.45 (moderate positive)
- **Peak Age:** 30-32 (maximum overall rating)
- **Insight:** Players improve until 30, then slight decline

Nationality Insights

- **Countries with Highest Avg Rating:** France, Spain, Germany, England
- **Highest-rated players:** Often from major European leagues

Position vs Overall Rating

- **Highest Avg Positions:** Attacking positions (ST, CF, LW, RW)

- **Lowest Avg Positions:** Goalkeeper (specialized skill set)

Market Value vs Overall Rating

- **Correlation:** +0.58 (moderate positive)
- **Driver:** Players with higher ratings command higher transfer values

Wage vs Overall Rating & Market Value

- **Overall → Wage:** +0.63 correlation
- **Value → Wage:** +0.81 correlation (strongest)
- **Insight:** Top earners are established superstars

Left-footed vs Right-footed Players

- **Left-footed Avg Rating:** Slightly higher (~70.2)
- **Right-footed Avg Rating:** ~69.8
- **Interpretation:** Rarity of left-foot gives marginal advantage

Multivariate Analysis

Key Correlations (Heatmap Analysis)

Relationship	Correlation Insight	
Value ↔ Wage	0.81	Market value strongly predicts wages
Special ↔ Crossing	0.86	Technical ability links to crossing
Skill Moves ↔ Finishing	0.73	Advanced skills correlate with finishing
Height ↔ Weight	0.76	Physical build consistency
Height ↔ Crossing	-0.50	Shorter players excel in crossing
Height ↔ Skill Moves	-0.44	Shorter players more agile
Age ↔ Potential	-0.32	Younger players have growth potential
Overall ↔ Potential	0.58	High ratings → high potential
International Rep ↔ Wage	0.61	Fame drives salary

Relationship

Correlation Insight

International Rep ↔ Value **0.54**

Reputation increases market value

Feature Engineering

1. Age Grouping

Created Age Groups:

- **Teen (15-19):** Youth players
- **Early 20s (20-24):** Young professionals
- **Mid 20s (25-29):** Prime years
- **Early 30s (30-34):** Experienced veterans
- **Late 30s+ (35+):** Super veterans

Value: Captures non-linear age effects

2. Overall Rating Classification

Created Rating Categories:

- **Low (40-59):** Below-average professionals
- **Average (60-69):** Standard squad players
- **Good (70-79):** Regular starters
- **Very Good (80-89):** Elite players
- **World Class (90+):** Superstars

Value: Enables classification insights alongside regression

3. Position Categorization

Grouped positions into tactical roles:

- **Attack:** ST, CF, LF, RF, LS, RS, LW, RW (8 positions)
- **Midfield:** CAM, CM, LCM, RCM, CDM, LDM, RDM, LM, RM, LAM, RAM (11 positions)
- **Defense:** CB, LCB, RCB, LB, RB, LWB, RWB (7 positions)

- **Goalkeeper:** GK (1 position)
- **Other:** SUB, RES (2 positions)

Value: Simplifies position feature while preserving tactical context

4. Contract Duration

Created Feature: Contract_years

- **Calculation:** Contract Year - Joined Year
- **Correlation with Overall:** -0.08 (weak)
- **Insight:** Tenure doesn't strongly predict rating

5. Body Mass Index (BMI)

Formula:
$$BMI = \frac{\text{Weight(kg)}}{(\text{Height(m)})^2}$$

Created Fitness Levels:

- **Underweight:** BMI < 18.5 (rare for athletes)
- **Fit:** $18.5 \leq \text{BMI} < 25$ (ideal athlete range)
- **Overweight:** $25 \leq \text{BMI} < 30$ (some strength players)
- **Obese:** BMI ≥ 30 (very rare for professional players)

Value: Captures physical conditioning proxy

6. Work Rate Decomposition

Split Work Rate into Two Features:

- **Attacking Work Rate:** Low/Medium/High intensity in attack
- **Defensive Work Rate:** Low/Medium/High intensity in defense

Value: Enables separate modeling of attacking vs defensive tendencies

Final Feature Set

- **Numerical Features:** 16 (including engineered)
 - **Categorical Features:** 13 (including engineered)
 - **Total Features Used:** 29
-

Machine Learning Models

Data Preparation

Train-Test Split: 80-20 split (80% training, 20% testing)

- Training samples: 13,368
- Testing samples: 3,342

Feature Preprocessing Pipeline:

ColumnTransformer:

- ├— OrdinalEncoder → Preferred Foot, Attacking/Defensive Work Rate, Fitness Level
- ├— OneHotEncoder → Position (creates binary columns for each position)
- └— StandardScaler → All numerical features (normalize to mean=0, std=1)

15+ Models Implemented

Linear Models

1. Linear Regression

- Simple baseline model
- R^2 Score: ~0.89
- RMSE: ~1.4

2. Ridge Regression

- L2 regularization (penalizes large coefficients)
- Handles multicollinearity
- R^2 Score: ~0.89
- RMSE: ~1.4

3. Lasso Regression

- L1 regularization (feature selection via coefficient shrinkage)
- Can zero out less important features
- R^2 Score: ~0.88
- RMSE: ~1.5

Distance-based Models

4. K-Nearest Neighbors (KNN)

- k=5 neighbors
- Non-parametric approach
- R^2 Score: ~0.88
- RMSE: ~1.5

Support Vector Machines

5. Support Vector Regressor (SVR)

- RBF (Radial Basis Function) kernel
- C=10 regularization parameter
- R^2 Score: ~0.87
- RMSE: ~1.6

Tree-based Models

6. Decision Tree Regressor

- max_depth=5 (prevents overfitting)
- Criterion: squared_error
- R^2 Score: ~0.88
- RMSE: ~1.5

7. Random Forest Regressor

- n_estimators=200 (200 trees)
- max_depth=10
- Ensemble of decision trees
- R^2 Score: **0.92** ✓
- RMSE: **1.2** ✓

8. Extra Trees Regressor (Extremely Randomized Trees)

- n_estimators=200

- max_depth=10
- Similar to Random Forest with more randomization
- R^2 Score: **0.92** ✓
- RMSE: **1.2** ✓

Boosting Models (Sequential Ensemble)

9. Gradient Boosting Regressor

- n_estimators=300 (iterations)
- learning_rate=0.1
- max_depth=5
- Sequential tree construction
- R^2 Score: **0.93** ✓✓
- RMSE: **1.1** ✓✓

10. Histogram-based Gradient Boosting

- Faster variant using binned data
- max_iter=100
- R^2 Score: **0.93** ✓✓
- RMSE: **1.1** ✓✓

11. AdaBoost Regressor

- n_estimators=100
- learning_rate=0.1
- Focuses on difficult samples
- R^2 Score: ~0.87
- RMSE: ~1.6

Averaging Ensemble

12. Bagging Regressor

- n_estimators=100

- Parallel tree construction
- R^2 Score: ~ 0.91
- RMSE: ~ 1.3

Stacking Models (Meta-learning)

13. Voting Regressor

- Combines: Linear Regression, Random Forest, Decision Tree, SVR
- Averaging ensemble approach
- R^2 Score: ~ 0.91
- RMSE: ~ 1.3

14. Stacking Regressor

- Base learners: Linear Regression, Random Forest, Decision Tree, SVR
- Meta-learner: XGBoost
- Advanced ensemble technique
- R^2 Score: **0.93** ✓✓
- RMSE: **1.1** ✓✓

Naive Bayes

15. Bayesian Ridge Regression

- Probabilistic approach with uncertainty estimation
- R^2 Score: ~ 0.89
- RMSE: ~ 1.4

Deep Learning

16. Fully Connected Neural Network (FCNN)

- Architecture:
 - Input Layer: 43 neurons (preprocessed features)
 - Dense (256) $\rightarrow$ BatchNorm $\rightarrow$ Dropout(0.3)
 - Dense (128) $\rightarrow$ BatchNorm $\rightarrow$ Dropout(0.2)

- Dense (64) → Dropout(0.1)
 - Dense (32) → Activation
 - Output (1) → Linear
- Training Details:
 - Optimizer: Adam (lr=0.001)
 - Loss Function: Mean Squared Error (MSE)
 - Callbacks: Early Stopping (patience=100), Model Checkpoint, CSV Logger
 - Epochs: 150 maximum
 - Batch Size: 32
 - Validation Split: 20%
- Performance:
 - R^2 Score: ~0.90
 - RMSE: ~1.3

Hyperparameter Optimization

Grid Search on Gradient Boosting:

Tuned parameters:

- n_estimators: [100, 200, 300, 400]
- learning_rate: [0.01, 0.05, 0.1]
- max_depth: [3, 5]

Cross-validation: 3-fold CV

Optimal Parameters Found:

- Best n_estimators: 300
 - Best learning_rate: 0.1
 - Best max_depth: 5
 - Best CV R^2 Score: ~0.93
-

Model Performance & Results

Top 5 Performing Models

Rank	Model	R ² Score	RMSE	MAE
1	Stacking Regressor	0.934	1.08	0.82
2	Gradient Boosting (Optimized)	0.933	1.10	0.83
3	Histogram GradientBoosting	0.932	1.11	0.84
4	Random Forest	0.921	1.23	0.93
5	Extra Trees	0.919	1.26	0.95

Metrics Interpretation

R² Score (Coefficient of Determination):

- Ranges from 0 to 1 (1 = perfect fit)
- **0.93:** Model explains 93% of overall rating variance
- Excellent performance

RMSE (Root Mean Squared Error):

- Average prediction error magnitude
- **1.1 rating points:** Very accurate (within ± 1 overall rating)

MAE (Mean Absolute Error):

- Average absolute deviation
- **0.82 rating points:** Typical prediction is within 0.82 points

Model Comparison Insights

1. **Ensemble & Boosting Methods Dominate:** Tree-based ensembles significantly outperform linear models
2. **Linear Models Plateau:** All linear models ~ 0.88 - 0.89 R²
3. **Gradient Boosting Sweet Spot:** Outperforms Random Forest, on par with Stacking
4. **Deep Learning Competitive:** FCNN achieves ~ 0.90 R², good but not best

5. **Hyperparameter Tuning Effective:** Grid search improved GB model $\sim 1.5\text{-}2\%$ in R^2

Key Insights

Player Characteristics

1. Age Matters Most:

- Players peak in rating around age 30-32
- Decline begins after 35
- Age stronger predictor than years at club

2. Position-Specific Patterns:

- Attacking positions: Highest average rating
- Defensive positions: Mid-range ratings
- Goalkeepers: Specialized, separate rating logic
- Midfielders: Balance between attack/defense

3. Physical Attributes:

- Height negatively correlates with agility (-0.44)
- Shorter players excel in dribbling/crossing
- Weight naturally follows height (+0.76)
- BMI relatively stable across ratings

4. Technical Skill Hierarchy:

- Ball Control, Dribbling, Passing drive overall rating
- Special attribute strongly links to crossing (0.86)
- Skill Moves indicate attacking prowess

Financial Relationships

1. Wage-Value Link (0.81):

- Market value is primary wage driver
- High R^2 in wage prediction

2. International Reputation Effect:

- Increases salary (0.61 correlation)
- Increases market value (0.54 correlation)
- Star players earn premium

3. Age-Value Relationship:

- Peak value: ages 27-32
- Young players: high potential but lower value
- Veterans: diminishing value after 34

Predictive Patterns

1. Feature Importance (from models):

- Age: ~15-18% importance
- Potential: ~12-14% importance
- Technical Skills (Ball Control, Passing): ~10-12% each
- Position: ~8-10% importance
- Physical Attributes: ~3-5% each

2. Non-linear Relationships:

- Age effect is non-linear (not constant)
- Position interacts with other features
- Skills have diminishing returns

3. Data Quality:

- Minimal noise despite 16,710 samples
- Outliers are valid (real superstars)
- Missing data well-handled through domain-aware imputation

Deployment

Model Serialization

Trained Model Saved: player_overall_model.pkl

- Format: Pickle (joblib)
- Contains: Full pipeline (preprocessing + Gradient Boosting model)
- Size: ~50MB

Deployment Code Example

```
import joblib
```

```
import pandas as pd
```

```
# Load model
```

```
model = joblib.load('player_overall_model.pkl')
```

```
# Prepare player data (same preprocessing applied)
```

```
player_data = pd.DataFrame({
```

```
    'Age': [28],
```

```
    'Potential': [88],
```

```
    'Value': [75000000],
```

```
    'Wage': [350000],
```

```
    'Special': [85],
```

```
    'International Reputation': [4],
```

```
    'Skill Moves': [5],
```

```
    'Height': [185],
```

```
    'Weight': [78],
```

```
    'Crossing': [85],
```

```
    'Finishing': [87],
```

```
    'SprintSpeed': [89],
```

```
    # ... other features
```

```
})
```

```
# Predict
```

```
predicted_overall = model.predict(player_data)
```

```
print(f"Predicted Overall Rating: {predicted_overall[0]:.1f}")
```

Production Considerations

1. API Integration:

- Wrap model in Flask/FastAPI application
- Accept player stats, return rating prediction
- Include confidence intervals from ensemble variance

2. Monitoring:

- Track prediction errors over time
- Detect model drift (updates to FIFA ratings)
- Re-train quarterly with new player data

3. Scalability:

- Model handles batch predictions efficiently
- Preprocessing scales to millions of records
- Typical prediction: <10ms per player

Project Structure

FIFA22_Player_Prediction/

├── 1785153964202_Player_Overall_Prediction.ipynb

├── FIFA22_official_data.csv (input)

├── player_overall_model.pkl (output)

├── training_log.csv (neural network logs)

└── best_model.keras (neural network weights)

└ Documentation/

└ player_prediction_documentation.md (this file)

🔧 Technologies & Libraries

- **Data Processing:** Pandas, NumPy
 - **Visualization:** Matplotlib, Seaborn, Plotly
 - **ML Models:** Scikit-learn (sklearn)
 - **Ensemble Methods:** XGBoost
 - **Deep Learning:** TensorFlow/Keras
 - **Missing Data:** missingno
 - **Model Serialization:** joblib
 - **Utilities:** re (regex), colour
-

Model Selection Recommendation

For Production Use: Gradient Boosting Regressor (Optimized)

Reasons:

1. ✓ Highest R^2 after hyperparameter tuning (0.933)
2. ✓ Fast prediction time (<5ms)
3. ✓ Robust to outliers and non-linear relationships
4. ✓ Feature importance interpretability
5. ✓ Smaller memory footprint than Stacking/FCNN
6. ✓ Handles missing data naturally
7. ✓ Retraining efficiency

Alternative (Highest Performance): Stacking Regressor

- Highest R^2 (0.934) but slower inference (~50ms)
- Better for accuracy-critical applications

🎓 Lessons & Best Practices Applied

1. **Exploratory Analysis First:** Understanding data distribution informed preprocessing choices
 2. **Domain Knowledge:** Sports context guided age grouping, position categorization
 3. **Outlier Retention:** Kept valid outliers; used robust models rather than deletion
 4. **Feature Engineering:** Created business-meaningful features (age groups, roles)
 5. **Pipeline Approach:** Ensured consistent preprocessing for train/test data
 6. **Ensemble Methods:** Combined multiple model types for superior performance
 7. **Hyperparameter Tuning:** Grid search significantly improved model accuracy
 8. **Validation Strategy:** Proper train-test split prevented overfitting assessment errors
-

✅ Conclusion

This project successfully demonstrates end-to-end machine learning workflow:

- **Clean Data:** 16,710 player records with proper handling of missing values
- **Thorough EDA:** Identified patterns in player attributes, ratings, and financial metrics
- **Feature Engineering:** Created 13 engineered features capturing domain knowledge
- **Model Diversity:** Implemented 16 different algorithms across categories
- **Superior Performance:** Achieved 93%+ R^2 score with <1.2 RMSE prediction error
- **Production Ready:** Serialized model ready for deployment with clear performance metrics

The **Gradient Boosting model** with optimized hyperparameters represents the optimal balance between accuracy, speed, interpretability, and maintainability for this prediction task.